

SOURABH CHAUHAN

Software Engineer | Python Developer | AI and ML Engineer | Data Engineer
TCS Python L3 Certified | Oracle Certified Generative AI Professional | Oracle Certified Data Science Professional
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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience designing and delivering enterprise software, intelligent automation, and AI-driven solutions across healthcare, banking, and financial services. Experienced in building scalable backend applications, cloud-native services, data processing pipelines, and machine learning solutions using Python, SQL, Docker, AWS, and modern DevOps practices. Proven track record of translating complex operational challenges into production-ready software that improves efficiency, reduces manual effort, and delivers measurable business impact.

EDUCATION

Master of Science in Computer Science
Bachelor of Computer Applications

Lakehead University, Canada | GPA: 4.0/4.0 | Aug 2024 - Sept 2025
Amity University, India | GPA 3.7/4.0 | Sept 2016 – 2019

PROFESSIONAL EXPERIENCE

AI Evaluation (Freelance) - Outlier AI

Nov 2025 – Jun 2026

- Evaluated large language model (LLM) responses for reasoning quality, factual accuracy, safety, instruction following, and policy compliance.
- Performed model analysis by identifying reasoning gaps, hallucinations, factual inaccuracies, formatting issues, and policy violations, while providing structured human feedback to improve model quality.

Tata Consultancy Services

March 2020 – Aug 2024

Enterprise clients: Cigna Healthcare, Express Scripts & Lloyds Bank Group.

Systems Engineer

April 2023 – Aug 2024

- Architected an AI-assisted virtual support platform using Python, NLP pipelines, REST APIs, and rule-based workflows to automate IT service requests, processing over 8,000 requests per month, reducing L1 support workload by 70%.
- Developed an enterprise-grade AI Knowledge Assistant using Python, FastAPI, LangChain, and Retrieval-Augmented Generation (RAG), implementing document ingestion, vector embeddings, semantic retrieval, and LLM-powered question answering to enable secure, context-aware access to enterprise knowledge.
- Designed and implemented an MLOps platform that processed infrastructure health logs every 12 hours using Isolation Forest, One-Class SVM, and a rule-based decision models to identify anomalies and automatically route diagnostic alerts.
- Engineered a centralized infrastructure diagnostics platform that automated server health validation, dependency checks, and patch compliance reporting across enterprise infrastructure, reducing manual operational reviews by 95%.
- Built containerized FastAPI microservices using Docker, GitHub Actions, CI/CD pipelines, and automated testing using Pytest, significantly improving deployment reliability.
- Collaborated with product owners, infrastructure engineers, and business stakeholders to translate operational challenges into production-ready solutions.

Asst. Systems Engineer

Feb 2021 – Mar 2023

- Developed machine learning pipeline using Python, XGBoost, and Logistic Regression models to classify incidents based on historical ticket data. Integrated predictions into ServiceNow to improve incident routing and reduce manual triage.
- Built SQL queries and PySpark ETL pipelines on AWS for extracting, validating, and transforming operational datasets supporting reporting, analytics, and automation initiatives.
- Developed reusable Python modules, improving maintainability, scalability, and software quality across multiple enterprise automation initiatives.
- Integrated ServiceNow, VMware, and Cisco UCS through Python APIs to automate diagnostics, regression checks, and operational workflows, reducing manual intervention while improving consistency across enterprise platforms.
- Delivered 12+ enterprise automation solutions across the full SDLC—from requirements analysis and architecture through deployment and production support—saving over 150 engineering hours of manual effort per month.

Automation Engineer

Mar 2020 – Jan 2021

- Automated Linux application deployments and module management using Python, Bash, Ansible, Docker, and CI/CD workflows, reducing manual deployment effort by over 90%.
- Engineered a Python orchestration platform using pyVmon and VMware APIs that automated virtual machine provisioning, configuration, and validation, reducing deployment time from 3 hours to under 10 minutes.
- Streamlined recurring infrastructure operations by automating user provisioning, snapshot lifecycle management, monitoring, and operational maintenance, reducing repetitive engineering effort by 80%.

- **Developed backend modules for enterprise HRMS and Home Services platforms** using C#, .NET, and SQL Server, implementing secure role-based access control, workflow automation, and database-driven business logic.
- **Implemented modules for employee access management, service booking workflows,** and user-role authorization systems.
- Supported **application testing, defect fixes, database updates, and technical documentation** during development and release cycles.

CORE SKILLS

- **Languages:** Python, SQL, Bash, PowerShell, YAML
- **Backend & APIs:** FastAPI, Flask, Django, REST APIs, Microservices, Object-Oriented Programming
- **AI & Machine Learning:** Scikit-learn, TensorFlow, XGBoost, Logistic Regression, Isolation Forest, KMeans, DBSCAN, NLP, CNNs, Image Processing, RAG, LangChain, Prompt Engineering
- **Data Engineering:** PySpark, Spark, Pandas, NumPy, Databricks, ELT/ETL Pipelines
- **Cloud & DevOps:** AWS (S3, EC2, Lambda), Docker, GitHub Actions, Terraform, CI/CD, Ansible, Airflow, Linux
- **Databases:** MySQL, PostgreSQL, Microsoft SQL Server, Oracle Database, MongoDB, SQLite
- **Testing:** Pytest, Unit Testing, Integration Testing, API Testing, Selenium, A/B Testing
- **Tools & Practices:** Git, Jira, Confluence, ServiceNow, Agile, Scrum, ITIL

ACADEMIC PROJECTS

- **Semantic Search Analytics Platform** | Python, Whisper, Flask, NLP, Scikit-learn | [Link](#)
 - Built an AI-powered semantic video search platform that transcribed videos, extracted contextual keywords using NLP techniques, and enabled topic-level search through Flask-based APIs.
- **Tic-Tac-Toe Game with Reinforcement Learning** | Python, Q-learning, Tkinter | [Link](#)
 - Engineered a self-learning agent using Q-learning - achieving over a 98%-win rate after 10,000 training episodes.
 - Implemented a Tkinter-based GUI to visualize gameplay and learning progress in real time.
- **AI Nutrition Analysis Platform** | Python, OCR, Computer Vision, Streamlit | [Link](#)
 - Developed an AI-based ingredient analysis platform that extracted product label data from images and generated health-risk insights using OCR, external nutrition datasets, and computer vision workflows.
- **Driver Alertness Detection App** | Python, Computer Vision, TensorFlow, OpenCV | [Link](#)
 - Built a real-time driver drowsiness detection system using TensorFlow, OpenCV, and computer vision techniques to detect driver drowsiness and fatigue using behavioral cues (e.g. droopy eyes, yawning).
 - Triggered visual and audio alerts to prompt rest, improving road safety and reducing accident risk.

AWARDS & RECOGNITION

- **Star of the Month Award** — Tata Consultancy Services
- **On-the-Spot Team Excellence Award** for enterprise automation delivery
- **Client Recognition Awards** for improving operational efficiency across enterprise environments.
- **Special Initiative and Contextual Master Awards** for introducing AI and machine learning–based automation and CI/CD pipeline improvements for client delivery teams

CERTIFICATIONS

- Oracle Certified Generative AI Professional [Link](#)
- Certified Data Science Professional [Link](#)
- Oracle Certified DevOps Professional [Link](#)
- IBM Data Analysis with Python | [Link](#)
- MongoDB for SQL Experts | [Link](#)
- **LangChain Academy:** Introduction to LangChain – Python | [Link](#)